

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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ABSTRACT. A report may specify a physical transformation without revealing the transformed state. We prove a causal energy identity for finite predictive test spaces transported by state-independent reports. At every finite-state update, information is lost twice: within each incoming row and when incoming conditional means are pooled. The identity gives a lower bound for all Borel randomized observers and an exact compatibility criterion for attaining its product bound. For compact-group relative transport with explicit orbital mass, global cover and stabilizer certificates, the optimal causal excess is comparable to the sum of the per-stage quantization scales, truncated at one. Spherical and higher-dimensional pure-state quantum orbits verify those certificates. On the circle the bound is exact. With an internally enforced deadline, every reachable phase is charged; balanced integer allocation gives the exact total-state resource curve. Positive classical transport and noncommuting quantum instruments verify the same theorem directly. Flagged hidden transformations yield a spherical-atomic acquired law with its true surviving mass and changing predictive rank. We also prove a fixed-start accessibility transfer for marked occupation certificates, explain cancellation without observable access, and compose same-task simulator and finite-precision errors without identifying labels with total bit complexity.

1. THE CAUSAL COST OF ACQUIRED GEOMETRY

The basic object is a performed experiment, not a parameter manifold. Preparation and positive causal kernels determine the executable future tests. Those tests determine a predictive quotient; a legal observer must update a finite representation of that quotient using only reports already delivered. The law actually obtained by this procedure, rather than a formal reachable set, supplies the geometric measure relevant to risk. A comparison of experiments must in addition implement the reports before their deadlines and pay for the simulator's retained state.

The difficulty addressed here is spatial and causal. An optimal quantizer of each acquired law need not be recursively executable. Even when the acquired law is identical at every time, repeated observation can force a new information loss at every finite-state update. This obstruction is different from a small probability of reaching a useful region or a long physical return time. It persists when the full-history update is an invertible isometry and every report has the same state-independent distribution.

Our central theorem identifies the obstruction at the level of conditional means of executable tests. Its exact identity separates the ordinary quantization loss of each incoming row from the additional loss caused by merging those rows into one next-state label. The latter term vanishes precisely when the incoming conditional centroids agree. Thus the same formula provides both an all-program lower bound and the compatibility condition required by its attaining causal construction. It is not an assertion that an arbitrary prescribed codebook can be updated for free.

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For reported isotropic transport on a predictive sphere of dimension d , let m_t be the number of labels retained after call t . The theorem and its geometric consequences give

$$\mathcal{R}_{\text{seq}}(n; m_1, \dots, m_n) - B \asymp_d \min \left\{ 1, \sum_{t=1}^n m_t^{-2/d} \right\}, \quad \mathcal{R}_{\text{cp}}(n, m) - B \asymp_d m^{-2/d}.$$

The constants are independent of n and all the m_t . The report reveals a relative transformation, not the current direction; no intermediate physical preparation occurs. The full-history baseline B belongs to the same terminal audit. This is a tracking experiment: n is a prescribed physical exposure, not an interchangeable number of independent samples of a static parameter.

On the circle, put $\beta_m = (\sin(\pi/m)/(\pi/m))^2$ for $m \geq 2$ and $\beta_1 = 0$. We obtain the exact sequential value $B + 1 - \prod_t \beta_{m_t}$ for arbitrary Borel randomized observers, not only for a quantizer architecture. If the observer must stop by its own state exactly after n reports, its initial state and all reachable phases are disjoint and charged. Writing $M - 1 = nq + r$, $0 \leq r < n$, gives the exact value

$$B + 1 - \beta_q^{n-r} \beta_{q+1}^r, \quad M \geq n + 1.$$

There is no legal observer for $M < n + 1$. In particular, feasibility is distinct from retaining any nonzero predictive energy. The optimal allocation is obtained by one integer division, rather than exhaustive synthesis.

The main proof starts from report instruments and finite test pullbacks. Sphere and circle calculations occur only after that theorem. Classical word-transport and noncommuting quantum instruments then verify its hypotheses independently. The homogeneous version includes higher-dimensional projective orbits, not only spheres. A flagged hidden-transform extension produces a true acquired mixture of a sphere and an atom, retaining the surviving mass in both the baseline and excess. A separate accessibility proposition converts inherited all-row physical-occupation certificates to a prescribed-start, observably gated task; a stationary mixture shows why this qualification cannot be removed.

Conditional projection, quantization and centroid identities are classical tools. We do not claim their invention. The contribution proposed here is the all-Borel causal product converse, its exact pooling compatibility, and the complete-state deadline law for relative rather than absolute observations. Section 10 compares the information pattern and conclusion with zero-delay coding and finite-memory partially observed control. The earlier marked physical-time theorem remains a distinct complete result, with its duration–exit dependence, prefix normalization and actual–return remainder; it is neither used to prove the new energy theorem nor discarded in favor of a model-specific calculation.

The higher-dimensional quantum realization derives the acquired dimension $2(D - 1)$ from Haar instruments and their executable audits, rather than substituting the ambient $D^2 - 1$ real matrix coordinates. Its mass lower is uniform over every possible incoming mixed conditional mean, including changes of eigenvalue multiplicity.

2. EXPERIMENTS AND EXECUTABLE PREDICTIVE COORDINATES

2.1. The physical interface. Preparation, hidden physical states, actions and reports are standard Borel, with normalized nonnegative causal kernels. Positivity means positivity preserving, not a lower bound on every report probability. A preparation is a paid physical operation. A report acquisition and a terminal audit are also paid operations. In the relative-transport experiment there are $n \geq 1$ report acquisitions between one preparation and one audit, hence $N = n + 2$ physical calls. The observer must commit its response before the audit. The environment-owned audit is not feedback, cannot be replaced by a simulator, and supplies no clock before that deadline.

The sequential model declares phase-dependent alphabets S_t , $|S_t| \leq m_t$, and Borel update probabilities $k_t(j \mid s, y)$. This model is externally scheduled; it is used first to identify the local

causal cost. The autonomous model instead has one finite set S , one reused Borel update row, an initial state, and a halt/readout map depending only on the retained state. It must halt exactly after n report acquisitions with probability one. There is no free phase, history tape, real-valued posterior register or persistent random seed. Fresh randomness may be used for the current update; any part surviving to the next call belongs to the retained label. The terminal readout is a fixed function of the final retained label, with values in the declared compact response set. A transient real-valued answer wire surviving until the audit is not a free extra register. Its real-number representation, program description, temporary workspace and time are distinct resources.

2.2. Raw finite test closure. Let $\mathcal{V}$ be a finite-dimensional real space spanned by the constant test and finitely many centered, bounded, executable future test evaluations. The space is constructed by test pullbacks of the raw kernels, not by prescribing a covariance or Fisher matrix. Coordinates in $\mathcal{V}/\mathbb{R}\mathbf{1}$ are selected once and their Euclidean norm is fixed by the squared task. Write $p(h) \in \mathbb{R}^k$ for the conditional predictions after a visible history h . The physical states need not be finite-dimensional.

The transport hypothesis is the following identity on this test space. Reports have a fixed probability law λ , independent of the entering physical state and of all earlier reports. The positive instrument at report y preserves constants and pulls back the centered tests by a Borel matrix $A(y)$, so that

$$(1) \quad \mathbb{E}[p_t \mid \mathcal{H}_{t-1}, Y_t = y] = A(y)p_{t-1}, \quad p_t = A(Y_t)p_{t-1}.$$

The second equality concerns the full visible history and follows when these tests close under the reported instrument. We require both facts as raw test identities; a state-dependent likelihood generally does not satisfy them. Stage-dependent matrices and reference laws are allowed. Throughout the general energy theorem assume

$$(2) \quad \int A_t(y)^T A_t(y) \lambda_t(dy) = \kappa_t I, \quad 0 < \kappa_t < \infty,$$

and bounded square moments. The normalized zero case can be treated directly and is excluded from division by κ_t . These conditions are checked by integration on the primitive report space. They are not assumptions about an observer's proposed quantizer.

There is a performed bounded terminal vector Z with $\mathbb{E}[Z \mid \mathcal{H}_n] = p_n$. Its second moment is determined by the preparation and kernels. Under squared loss, for any retained-state response $a(S_n)$,

$$(3) \quad \mathbb{E} \|Z - a(S_n)\|^2 = B_n + \mathbb{E} \|p_n - a(S_n)\|^2, \quad B_n = \mathbb{E} \|Z\|^2 - \mathbb{E} \|p_n\|^2.$$

The same B_n is used for checkpoint and online comparisons. In the orbit experiments $\|p_n\| = 1$ and $\mathbb{E} \|Z\|^2 = V$, hence $B = V - 1$. Projecting a response into the closed unit ball does not increase loss to a predictive point in that ball. The response coordinates may be unbiased estimators assembled from a randomly selected bounded physical test; they need not be jointly measured incompatible observables.

Lemma 2.1 (Realized predictive quotient). *Suppose the prescribed executable tests have conditional evaluations affine in $p(h)$, contain the coordinate tests, and include their legal finite continuations. If p is Borel with standard-Borel image and a Borel section, equality of p , together with visible interface coordinates, is precisely predictive equivalence for this test family. The quotient is minimal among Borel statistics carrying all these test predictions. Fiber-invariant report and update kernels, jointly Borel in any legal action, descend to it.*

Proof. Equality of the coordinates implies equality of every affine test evaluation; the converse follows from the included coordinate tests. Any Borel statistic carrying them factors p coordinatewise. Evaluation at a Borel section defines the descended kernels and fiber invariance makes their value

independent of the section. Joint action measurability is retained by composition. Legal phases and available budgets are not identified merely because the numerical coordinates coincide. $\square$

For the raw orbit realizations, the image is the sphere, or the sphere together with its center after a hidden transform. A Borel rotation sending a fixed unit vector to a prescribed direction can be selected on finitely many coordinate charts. A one-call visible history therefore supplies a section. The full future experiment consists of relative instruments and the declared linear audit tests, so these coordinates are its predictive quotient, not an asserted quotient of additional unmodeled physical tasks. The general result permits a task factor when other tests are explicitly left outside this family.

2.3. The acquired orbit. In the isotropic case $A(y) = R(y)$ is orthogonal, and the report law pushes forward to Haar probability on $SO(k)$, with $k = d + 1 \geq 2$. The prepared predictive point $p_0 = x_*$ is a known unit vector. At each call

$$(4) \quad p_t = R_t p_{t-1}, \quad R_1, \dots, R_n \text{ independent Haar.}$$

Thus the actual acquired law of p_t is uniform sphere measure σ_d for every $t \geq 1$. This law is derived from the performed reports. Conditional on a report the map is invertible and no preparation restarts the prediction. The unconditional transition being isotropic does not give the observer an absolute-state report.

Additional declared actions may apply known orthogonal transformations before the independent Haar report. The proof conditions on the current action as well as the old label, and Haar invariance removes that transformation. It covers Borel randomized choices made from the counted label and fresh randomness among these actions. A controller with access to additional uncharged history, or sensing actions that directly reveal p_t , is not included.

3. A CAUSAL ENERGY AND COMPATIBILITY THEOREM

For a finite measure ν with finite second moment on $\mathbb{R}^k$ define

$$q_m(\nu) = \inf_{c_1, \dots, c_m} \int \min_j \|x - c_j\|^2 \nu(dx).$$

The measure is not normalized if it is a subprobability. For probability measures, conditional projection gives

$$(5) \quad \sup_{J: |J| \leq m} \mathbb{E} \|\mathbb{E}[X \mid J]\|^2 = \mathbb{E} \|X\|^2 - q_m(\text{Law } X),$$

where randomized Borel encoders are included. Indeed, for any output means the randomized assignment is no better than nearest-center assignment, and optimizing the means gives the reverse inequality. For compact support, centers lie in its closed convex hull and compactness gives attainment. The identity with infima needs only a finite second moment, by truncation or the same nearest-center inequalities. This is the standard squared quantization identity [1].

Let $v_{t,s} = \mathbb{E}[p_t \mid S_t = s]$, define values arbitrarily on zero-mass labels, and put $e_t = \mathbb{E} \|v_{t,S_t}\|^2$. A label is not a free copy of p_t : its prediction is its conditional mean under its own actual law. Given $S_{t-1} = s$, let

$$W_s = A_t(Y_t)v_{t-1,s}, \quad c_{s,j} = \mathbb{E}[W_s \mid S_{t-1} = s, S_t = j].$$

The update randomness is included in this conditional law. Let $\pi_s = \mathbb{P}(S_{t-1} = s)$ and $\pi_{s,j} = \mathbb{P}(S_{t-1} = s, S_t = j)$.

Theorem 3.1 (Causal conditional-mean energy transfer). *Under the raw test and report hypotheses of Section 2, every finite-state Borel randomized observer satisfies the exact identity*

$$(6) \quad e_t = \kappa_t e_{t-1} - D_t^{\text{row}} - D_t^{\text{pool}},$$

where

$$(7) \quad D_t^{\text{row}} = \sum_s \pi_s \mathbb{E}[\|W_s - c_{s,S_t}\|^2 \mid S_{t-1} = s],$$

$$(8) \quad D_t^{\text{pool}} = \sum_{s,j} \pi_{s,j} \|c_{s,j} - v_{t,j}\|^2.$$

In particular, with

$$(9) \quad \eta_t(m) = \inf_{\|v\|=1} \frac{q_m(\text{Law}(A_t(Y_t)v))}{\kappa_t},$$

one has

$$(10) \quad e_n \leq e_0 \prod_{t=1}^n \kappa_t (1 - \eta_t(m_t)).$$

The bound holds for each legal observer, not just a chosen quantizer. Equality in a nonzero one-step bound holds if and only if every positive-mass incoming row with nonzero entering mean attains the coefficient in (9), its encoding attains its corresponding q_{m_t} , and all incoming centroids with the same outgoing label agree. Thus compatible conditional centroids are necessary and sufficient to attain the local product bound. Successive compatible rows attain the full product whenever its factors and entering energies are nonzero.

For isotropic relative transport on $\mathbb{S}^d$, set $\beta_d(m) = 1 - q_m(\sigma_d)$. Then

$$(11) \quad \mathcal{R}_{\text{seq}}(n; m_1, \dots, m_n) - B \geq 1 - \prod_{t=1}^n \beta_d(m_t), \quad \mathcal{R}_{\text{cp}}(n, m) - B = q_m(\sigma_d).$$

A compatible centroid construction attaining the product is therefore an exact causal realization, not merely a checkpoint codebook.

Proof. The report is independent of the past and the update sees the old label and current report, with fresh randomness only. Hence

$$\mathbb{E}[p_t \mid S_{t-1}, Y_t, S_t] = A_t(Y_t)v_{t-1,S_{t-1}}.$$

Taking its conditional expectation given (S_{t-1}, S_t) gives $c_{s,j}$, and conditioning again on $S_t = j$ gives $v_{t,j}$. Applying the elementary conditional-square decomposition first inside each row and then when pooling rows gives

$$\sum_s \pi_s \mathbb{E} \|W_s\|^2 = D_t^{\text{row}} + \sum_{s,j} \pi_{s,j} \|c_{s,j}\|^2 = D_t^{\text{row}} + D_t^{\text{pool}} + e_t.$$

The left side is $\kappa_t e_{t-1}$ by (2). For each nonzero $v_{t-1,s}$, identity (5) and homogeneity bound the row loss below by $\kappa_t \eta_t(m_t) \|v_{t-1,s}\|^2$. A zero vector contributes zero. Since both losses are nonnegative, this proves the one-step inequality and its iteration. Equality requires equality in every positive-weight nonnegative term. The pooling term vanishes exactly when $c_{s,j} = v_{t,j}$ for every positive $\pi_{s,j}$; these observations prove the stated criterion, including its converse. If a factor or energy is zero, the product bound is zero and there is no inference about earlier equality; we make no such inference.

For Haar rotations, $R_t v$ is uniform on the sphere of radius $\|v\|$, including every direction of every incoming conditional mean. Thus $\kappa_t = 1$ and $\eta_t(m) = q_m(\sigma_d)$, not a coefficient involving the ambient physical dimension. Also $e_0 = 1$. The optimal terminal response is v_{n,S_n} , and its risk is $V - e_n$. Subtract $B = V - 1$ to obtain the first assertion in (11). A checkpoint sees the whole legal history, hence p_n , and can realize any Borel m -point quantizer of its actual law σ_d . This proves its identity. $\square$

The theorem is an exact finite-history statement; neither a regenerative boundary nor a contraction of the full-history update is assumed. It uses a finite test closure and state-independent report transport. General nonlinear filtering with state-dependent likelihoods is not silently included. An analytical conditional mean is used to prove the converse; the construction below stores only a label and never computes the exact hidden conditional distribution at runtime.

4. MATCHING GEOMETRIC LAWS WITHOUT RESET

The geometric consequence extends beyond spheres. Let G be a compact matrix group acting orthogonally on the centered raw test space V , with Haar reports and prepared unit test point x_* . Let $\mathcal{O} = Gx_*$ and $\mu = (g \mapsto gx_*)_{\#}h$. All points and measures below are induced by the performed instruments. Assume: (a) V has no nonzero G -fixed vector; (b) the stabilizer of x_* has fixed space exactly $\mathbb{R}x_*$ in V ; (c) for every unit $v \in V$, every Euclidean center z and $0 < r \leq r_0$,

$$h\{g : \|gv - z\| \leq r\} \leq C_0 r^d;$$

and (d) $\mathcal{O}$ has global m -point unit covers of radius at most $C_1 m^{-1/d}$. These are primitive orbital mass and cover certificates, uniform over all possible incoming conditional means; a bound only on the prepared orbit is not enough for the lower.

Theorem 4.1 (Homogeneous acquired-geometry transfer). *Under (a)–(d), for all $n \geq 1$ and $m_t \geq 1$,*

$$(12) \quad \mathcal{R}_{\text{seq}}(n; \mathbf{m}) - B \asymp \min \left(1, \sum_t m_t^{-2/d} \right), \quad \mathcal{R}_{\text{cp}}(n, m) - B = q_m(\mu) \asymp m^{-2/d}.$$

The infimum is over all Borel randomized label updates. An autonomous internal n -report deadline is feasible exactly for $M \geq n + 1$, and in that range

$$(13) \quad \mathcal{R}_{\text{aut}}(n, M) - B \asymp \min \left(1, n^{1+2/d} (M - 1)^{-2/d} \right).$$

Constants depend only on the displayed raw geometric certificates. Report laws have zero state-dependent total-variation distance, updates are isometries on the entire test space, and the stored observer uses global covers of $\mathcal{O}$. Thus stability, cover validity and report continuity are part of the proved implementation, not an assumed oracle filter.

Proof. Condition (c), with radius a fixed constant times $m^{-1/d}$, leaves at least half the actual orbital mass outside any m balls. Therefore $q_m(\text{Law}(gv)) \geq cm^{-2/d}$ uniformly in unit v , including all normalized incoming observer means. The energy theorem proves the lower product and hence the lower in (12). For a nearest-direction global cover Q_m on $\mathcal{O}$, let $\kappa_m = \int \langle u, Q_m(u) \rangle \mu(du)$. The zero Haar mean in (a) and the covering identity give $0 \leq \kappa_m \leq 1$ and $1 - \kappa_m \leq Cm^{-2/d}$.

For any $x, \hat{x} \in \mathcal{O}$, average gx conditional on $g\hat{x} = u$. Haar disintegration on the stabilizer coset and (b) give $\mathbb{E}[gx \mid g\hat{x} = u] = \langle x, \hat{x} \rangle u$. The recursive cover therefore multiplies the correlation by κ_{m_t} at each step. The legal terminal readout $(\prod_t \kappa_{m_t}) \hat{x}_n$ has predictive error $1 - \prod_t \kappa_{m_t}^2$, proving the upper. This readout lies in $\text{conv } \mathcal{O}$ because zero is its Haar barycenter. The checkpoint equality follows by conditioning and its two bounds follow from (c) at $v = x_*$ and (d). Finally the disjoint-phase proof of Lemma 5.3 applies to the same independent Haar report law on G . The budget $1 + \sum_t m_t \leq M$, Jensen's inequality and balanced integer widths prove (13), including the saturated range. No enumeration of histories enters the lower. $\square$

Lemma 4.2 (Acquired spherical mass and global covers). *For each integer $d \geq 1$ there are positive constants c_d, C_d such that, for all $m \geq 1$,*

$$(14) \quad c_d m^{-2/d} \leq q_m(\sigma_d) \leq C_d m^{-2/d}.$$

There are m or fewer unit directions and a Borel nearest-direction map Q_m with

$$\kappa_d(m) := \int \langle u, Q_m(u) \rangle \sigma_d(du) \geq 1 - C_d m^{-2/d}, \quad 0 \leq \kappa_d(m) \leq 1.$$

The cover is global on the sphere, not merely on the support of a selected candidate trajectory.

Proof. The spherical-coordinate area formula gives constants $c, C > 0$ such that

$$cr^d \leq \sigma_d(B(u, r)) \leq Cr^d \quad (u \in \mathbb{S}^d, 0 < r \leq 1).$$

The upper bound also holds, after increasing C , for Euclidean centers in the unit ball: a nonempty ball is contained in a spherical ball of twice its radius around a point of its intersection. For $r = (2Cm)^{-1/d}$, at least half the actual mass lies outside any m radius- r balls. This gives the lower bound. A maximal h -separated set is an h -cover and, by disjoint radius- $h/2$ balls, has at most $C'h^{-d}$ points. Choose h of order $m^{-1/d}$; boundedly many small values of m are absorbed in the constants. Nearest-direction assignment is Borel after breaking ties by index, and its squared distance is at most h^2 . Since $\|u\| = \|Q_m(u)\| = 1$, its mean inner product is at least $1 - h^2/2$. It is nonnegative because the maximum inner product among the chosen directions dominates any single direction, whose mean inner product is zero. The centroid codebook can only improve the distance upper, giving (14). $\square$

Theorem 4.3 (Nonreset acquired-geometry resource law). *For the raw isotropic transport interface and the same bounded terminal task, all integers $n \geq 1$ and $m_1, \dots, m_n \geq 1$ satisfy*

$$(15) \quad c_d \min \left(1, \sum_{t=1}^n m_t^{-2/d} \right) \leq \mathcal{R}_{\text{seq}}(n; m_1, \dots, m_n) - B \leq C_d \min \left(1, \sum_{t=1}^n m_t^{-2/d} \right).$$

The infimum is over all legal Borel randomized finite-state observers. The upper is attained to the displayed constants by explicit deterministic recursive covers. The checkpoint identity in (11) holds at the same horizon and task. Thus the acquired law alone does not determine the online value: the relative-report interface and its repeated compression loss are essential.

Proof. The lower follows from Theorem 3.1 and Lemma 4.2, using $1 - \prod_t (1 - a_t) \geq 1 - \exp(-\sum_t a_t)$ for $0 \leq a_t \leq 1$. For $d \geq 2$ store a unit code direction $\hat{p}_t$, initialized at x_* , and update

$$\hat{p}_t = Q_{m_t}(R_t \hat{p}_{t-1}).$$

Fix entering unit vectors $p, \hat{p}$. Conditional on $R\hat{p} = u$, the stabilizer $SO(d)$ averages every perpendicular component to zero. Therefore

$$\mathbb{E}[Rp \mid R\hat{p} = u] = \langle p, \hat{p} \rangle u,$$

and averaging over u gives

$$\mathbb{E}\langle p_t, \hat{p}_t \rangle = \kappa_d(m_t) \mathbb{E}\langle p_{t-1}, \hat{p}_{t-1} \rangle.$$

Writing $a_n = \prod_t \kappa_d(m_t)$, use the terminal response $a_n \hat{p}_n$, whose scalar is a precomputed program constant. Its squared predictive error is $1 + a_n^2 - 2a_n^2 = 1 - a_n^2$. It is at most $\sum_t (1 - \kappa_d(m_t)^2)$ and at most one, so the geometric bounds prove the upper. On the circle the regular-arc map has zero mean sine error and mean cosine error $\sin(\pi/m)/(\pi/m)$; direct rotation integration proves the same correlation identity. Section 5 also proves the exact value there. No full-history posterior is present in the executing recursion. $\square$

The full-history update has Lipschitz constant one, not a strict contraction. In the space of test predictions the report kernels are state independent, and every linear audit probability is Lipschitz. Cover validity and report continuity are therefore verified from the primitive interface. Error does not disappear between preparations, because there are no intermediate preparations. The matching law measures precisely its accumulation.

5. EXACT CAUSAL ALLOCATION ON THE CIRCLE

Write σ_1 for normalized arclength on the unit circle, and set

$$(16) \quad a_m = \frac{\sin(\pi/m)}{\pi/m}, \quad \beta_m = a_m^2 \quad (m \geq 2), \quad a_1 = \beta_1 = 0.$$

These are Euclidean squared-distortion coefficients with centroids allowed in the disk. They are not geodesic distortion coefficients or codepoints constrained to the circle.

Lemma 5.1 (Circle centroids). *For every $m \geq 1$, $q_m(\sigma_1) = 1 - \beta_m$. Equal arcs and their centroids attain the value, including randomized Borel encoders in the converse.*

Proof. For an encoder cell of mass p , allowing a fractional membership weight in $[0, 1]$, its first moment has norm at most $\sin(\pi p)/\pi$. To see this, test the moment in its own unit direction. Among functions of prescribed integral p , the integral of the cosine is maximized by putting weight one on the centered arc of length $2\pi p$; exchanging any lower cosine value inside with a larger one outside improves the integral. Direct integration gives the bound.

Put $f(p) = \sin^2(\pi p)/(\pi^2 p)$ and $f(0) = 0$. The total retained energy is at most $\sum_j f(p_j)$. The function is concave on $[0, 1/2]$: after writing $z = \pi p$, the sign of its second derivative is that of $h(z) = z^2 \cos(2z) - z \sin(2z) + \sin^2 z$; $h(0) = 0$ and $h'(z) = -2z^2 \sin(2z) \leq 0$ there. If $p > 1/2$ is one mass, any other positive mass q is at most $1 - p$. For $0 < u \leq 1/2$,

$$f'(u) - f'(1 - u) = \frac{\sin^2(\pi u)}{\pi^2 u^2 (1 - u)^2} [2\pi u(1 - u) \cot(\pi u) - (1 - 2u)] \geq 0.$$

Indeed $\cos(\pi u) \geq 1 - 2u$ by concavity and $\sin(\pi u) \leq \pi u$, so the bracket is nonnegative. Concavity on the half interval gives $f'(q) \geq f'(1 - p) \geq f'(p)$. Transferring mass from p to q until the larger mass is at most $1/2$ cannot decrease the sum. Zero masses are handled by continuity. Once all masses lie in $[0, 1/2]$, Jensen's inequality gives $\sum_j f(p_j) \leq m f(1/m) = \beta_m$ for $m \geq 2$. For $m = 1$ the only mean is zero. Equal arcs have masses $1/m$ and centroids $a_m e^{2\pi i j/m}$, and attain every bound. Use (5) to finish. $\square$

The static circle formula is established in the quantization literature [2]; the proof is included to specify the loss and allow arbitrary fractional cells. Our causal use requires the additional compatibility argument below.

Theorem 5.2 (Exact sequential product). *For all $n \geq 1$ and $m_t \geq 1$,*

$$(17) \quad \mathcal{R}_{\text{seq}}(n; m_1, \dots, m_n) = B + 1 - \prod_{t=1}^n \beta_{m_t}.$$

A deterministic recursive regular-arc encoder attains it. If all factors are positive, every attaining observer satisfies the local optimality and pooled-centroid conditions of Theorem 3.1 at every positive-mass transition.

Proof. The lower is Theorem 3.1 and Lemma 5.1. Suppose the old conditional means are r times known unit directions. Given an old label, rotate its direction by the report and assign the result to one of m_t equal arcs. Each outgoing label has probability $1/m_t$ in every incoming row. Its conditional mean is $r a_{m_t}$ times the corresponding new direction, independently of the old label. The row quantization is optimal and the incoming centroids agree exactly, so the pooling defect is zero. Induct from $r = 1$ at the known preparation to $r = \prod_t a_{m_t}$. The terminal conditional mean is the legal fixed readout of that label. If a factor is zero, thereafter the mean is zero and the zero-response observer attains the same value. The equality criterion when all factors are positive follows by tracing equality through the nonnegative losses. $\square$

Lemma 5.3 (The internal deadline consumes disjoint phases). *Suppose the observer has one reused Borel transition rule, its halt decision is a function of the retained label, and every report has the same Haar law independently of prior reports. If it halts exactly after $n \geq 1$ reports almost surely, the sets of labels occurring with positive probability after $0, 1, \dots, n$ reports are pairwise disjoint. If their sizes after the positive times are $m_1, \dots, m_n$, then*

$$(18) \quad 1 + \sum_{t=1}^n m_t \leq M.$$

Conversely any such phase alphabets can be implemented by one autonomous program with exactly that many labels.

Proof. Integrating a report row against Haar gives one time-homogeneous transition matrix on labels; halt labels have no further acquisition. If a nonhalt label occurs at two times $s < t < n$, its future stopping-time distribution, conditional on that label, is the same at both times. It must be concentrated at both $n - s$ and $n - t$, a contradiction. A halt label cannot occur before n , and a nonhalt label cannot occur at n . Initial randomization only increases the number of possible time-zero labels, which is at least one. This proves (18). For the converse use the disjoint union of the phase alphabets and a single initial label, with each phase row wired to the next and the last phase marked as halt. The report itself supplies no phase. Known pre-report orthogonal actions preserve the integrated Haar law and do not change the argument. $\square$

Theorem 5.4 (Exact total-state law and allocation). *For the circle experiment with an internal exact n -report deadline, the legal M -label class is nonempty exactly when $M \geq n + 1$. In that range write*

$$K = M - 1 = nq + r, \quad q \geq 1, \quad 0 \leq r < n.$$

Then the infimum over all autonomous Borel randomized programs is attained, and

$$(19) \quad \mathcal{R}_{\text{aut}}(n, M) = B + 1 - \beta_q^{n-r} \beta_{q+1}^r.$$

In particular, the excess equals one when $n + 1 \leq M \leq 2n$, and

$$(20) \quad \mathcal{R}_{\text{aut}}(n, M) - B \asymp \min\{1, n^3/(M - 1)^2\}.$$

More generally, for $\mathbb{S}^d$,

$$(21) \quad \mathcal{R}_{\text{aut}}(n, M) - B \asymp_d \min\{1, n^{1+2/d}/(M - 1)^{2/d}\}.$$

Preparation and audit add two physical calls, so these formulas use $n = N - 2$. No free-clock interpretation is made.

Proof. The clock lemma proves infeasibility below $n + 1$. It also partitions every legal observer into a sequential architecture satisfying (18). The product theorem bounds its energy by $\prod_t \beta_{m_t}$ and supplies a program attaining every such allocation. For real $x > 1$ the increasing function $\ell(x) = 2 \log(\sin(\pi/x)/(\pi/x))$ is strictly concave, because with $z = \pi/x$,

$$\ell''(x) = -\frac{2}{x^2} [(1 - z \cot z)^2 + z^2] < 0.$$

An exchange between two integer widths differing by at least two increases their product. Thus positive products are maximized by the balanced allocation $q, q + 1$. If $K < 2n$, every allocation has a width one and product zero, so the same formula remains valid without taking $\log \beta_1$. Widths can use all K labels since β_m increases. The explicit regular-arc construction proves attainment and the asserted threshold.

The rate (20) follows from $1 - \beta_m \asymp m^{-2}$ and the product bounds. For the sphere, Jensen gives $\sum_t m_t^{-2/d} \geq n^{1+2/d} K^{-2/d}$, while the balanced integer allocation has a sum at most $2^{2/d}$ times this quantity. Apply Theorem 4.3. These arguments include $q = 1$ by the truncation at one. $\square$

Corollary 5.5 (Resolution and synthesis). *For $0 < \rho \leq \rho_0 < 1$, excess at most ρ^2 on the sphere requires and, up to constants depending only on d, ρ_0 , is attained by*

$$M - 1 \asymp_d n^{1+d/2} \rho^{-d}.$$

A checkpoint requires $m \asymp_d \rho^{-d}$ instead. On the circle the optimal phase widths require one quotient and remainder computation on the binary integers $M - 1, n$; ordinary long division uses $O((\log M + \log n)^2)$ bit operations. This is an exact allocation result, not an efficient global synthesis claim for arbitrary kernels. Compiling all transition rows and their real constants has the separate cost stated in Proposition 9.2.

Proof. Solve (21) in the nonsaturated range, and use the checkpoint bound. The arithmetic claim follows from the balanced formula; enumerating a table of all states still costs at least its output length. $\square$

6. SINGULAR ACQUISITION WITH A GENUINE SURVIVING MASS

Consider the same relative transport, but at call t independently suppress the rotation report with probability $1 - \alpha_t$, $0 \leq \alpha_t \leq 1$. A suppressed report is the visible symbol $\perp$; the physical rotation is still performed and is not redrawn or undone. Conditional on $\perp$, its independent Haar average sends every entering test mean to zero. Subsequent reported rotations preserve this zero mean. Write

$$w_t = \prod_{j=1}^t \alpha_j.$$

The actual full-history predictive law is therefore

$$(22) \quad \nu_t = w_t \sigma_d + (1 - w_t) \delta_0.$$

This is a probability with a d -dimensional stratum and a zero-dimensional stratum; the first stratum is not renormalized. The actual full-history baseline is $B_t = V - w_t$.

Theorem 6.1 (Surviving-stratum causal law). *For all phase widths $m_t \geq 1$, every Borel randomized observer satisfies*

$$(23) \quad \mathcal{R}_{\text{seq}}^{\text{er}}(n; \mathbf{m}) - B_n \geq w_n \left(1 - \prod_t \beta_d(m_t) \right).$$

For $m_t \geq 2$ there is an observer with one counted inactive label in each phase and at most $m_t - 1$ active labels for which

$$(24) \quad \mathcal{R}_{\text{seq}}^{\text{er}}(n; \mathbf{m}) - B_n \leq C_d w_n \min \left\{ 1, \sum_t (m_t - 1)^{-2/d} \right\}.$$

On the circle its excess is exactly $w_n(1 - \prod_t \beta_{m_t-1})$ for this construction. Hence, for all $m_t \geq 2$, the optimal excess is comparable to $w_n \min(1, \sum_t m_t^{-2/d})$, uniformly even when $w_n \rightarrow 0$. The checkpoint excess lies between $w_n q_m(\sigma_d)$ and $w_n q_{m-1}(\sigma_d)$ for $m \geq 2$. For an autonomous exact deadline, all inactive phase labels are charged too; balanced widths give the analogous matching rate for $M \geq 2n + 1$ provided the report laws at different phases are mutually equivalent (for example $0 < \inf_t \alpha_t \leq \sup_t \alpha_t < 1$).

Proof. For an entering observer mean v_s , the conditional new mean before its label update is $R_t v_s$ on a disclosed report and zero on $\perp$. Its energy is $\alpha_t \|v_s\|^2$. The row distortion, for any m_t centers, is at least the contribution of the disclosed part, namely $\alpha_t \|v_s\|^2 q_{m_t}(\sigma_d)$. The energy identity, with the additional point mass at zero, gives $e_t \leq \alpha_t \beta_d(m_t) e_{t-1}$. Thus $e_n \leq w_n \prod_t \beta_d(m_t)$ and risk $V - e_n$ proves (23).

For the upper, move to the inactive label on the first $\perp$ and retain it thereafter, outputting zero. On the event of no suppression run the construction of Theorem 4.3 with $m_t - 1$ directions. Suppression is independent of the directions and the full active history, so this event has its actual mass w_n and the original conditional law. The inactive part has zero predictive error. This proves the upper, including the circle formula. Since $m_t - 1 \geq m_t/2$, the two bounds match to constants. A checkpoint may add the origin to an $(m - 1)$ -center codebook; restriction to the spherical submeasure gives its lower. For the autonomous construction use disjoint phases including their inactive labels and the initial label; the lower clock argument extends from equal laws to mutually equivalent phase laws. Indeed the integrated transition matrices then have the same positive-edge graph. A positive-probability path from a label to a halt label has the same possible length from every phase at which that label is reached, so the two different required remaining lengths are impossible. The autonomous upper itself needs no such lower clock argument. $\square$

The phase-law qualification in the last sentence matters. If a deterministic pattern of visible erasures encodes the phase, it supplies an external clock and the autonomous clock lower does not follow. The sequential and checkpoint statements hold for every $\alpha_t \in [0, 1]$ without that qualification. Also the baseline rises by $1 - w_n$ relative to the fully disclosed task: missing reports do not make the total risk smaller merely because the acquired sphere loses mass.

Corollary 6.2 (Homogeneous singular extension). *For the group experiments of Theorem 4.1, independent hidden reports give the actual law $w_n\mu + (1 - w_n)\delta_0$. For $m_t \geq 2$, the sequential excess over $V - w_n$ is comparable to $w_n \min(1, \sum_t m_t^{-2/d})$. With mutually equivalent phase report laws and $M \geq 2n + 1$, the autonomous excess is comparable to $w_n \min(1, n^{1+2/d}(M - 1)^{-2/d})$. The checkpoint bounds retain the factor w_n as above.*

Proof. Haar averaging is zero on V by (a). A hidden report thus annihilates the full-history test mean. For each incoming vector, the row distortion on the visible part is at least $\alpha_t c \|v\|^2 m_t^{-2/d}$ by (c). The energy identity gives the corresponding product lower. For the upper reserve one inactive label per phase and use the homogeneous recursive cover on the surviving event. Conditioning on that event recovers exactly the original group experiment, with mass w_n . The same phase-support and balanced-allocation arguments prove the autonomous assertion. No mass is renormalized in either direction. $\square$

Corollary 6.3 (One-task calibration and disclosure loss). *Append a terminal, environment-owned coordinate $s\delta$, where $s \in \{-1, 1\}$ is fixed but never reported and $0 \leq \delta \leq 1$. Under the sum of the original squared loss and the squared loss for this coordinate, minimax over s adds exactly δ^2 to all the displayed values. In particular, in the comparable sequential range,*

$$(25) \quad \mathcal{R}_\delta^{\text{er}} - B \asymp_d (1 - w_n) + w_n \min \left(1, \sum_t m_t^{-2/d} \right) + \delta^2.$$

This is a sharp selected resource region. It does not assert simultaneous optimality of program bits, workspace or execution time.

Proof. Use the equal prior on the two invisible signs for the lower: conditional on every legal history and random seed the conditional mean of $s\delta$ is zero, so that coordinate costs at least δ^2 . The identically zero response attains it without another label. The original terminal task and calibration coordinate are independent under this prior, so the losses add. Subtract $B = V - 1$ rather than the erased baseline $B_n = V - w_n$, and use Theorem 6.1. $\square$

7. TWO RAW POSITIVE REALIZATIONS

7.1. Classical relative transport. Fix $d \geq 1$, $k = d + 1$, a known point $x_* \in \mathbb{S}^d$, and a known contrast $0 < \eta \leq 1$. One raw realization has hidden physical state a finite word w of rotations,

in the standard Borel disjoint union $\bigsqcup_{j \geq 0} SO(k)^j$. Preparation creates the empty word. A call appends an independent Haar rotation and reports that rotation, but never allows rereading the word. Set $x(w)$ to its ordered product applied to x_* . The induced coordinate kernel is

$$(26) \quad K(dx', dR | x) = \delta_{Rx'}(dx') h(dR),$$

where h is Haar probability on $SO(k)$. Only R is reported. The audit chooses I uniformly from $\{1, \dots, k\}$ and then $S \in \{-1, 1\}$ with probability $(1 + S\eta x_I)/2$. Its scored vector is $Z = (k/\eta)Se_I$, so $\mathbb{E}[Z | x] = x$ and $\|Z\|^2 = k^2/\eta^2$. The audit happens after the answer has been committed and is not a further observation supplied to the machine.

Theorem 7.1 (Classical realization). *For the raw preparation, kernel (26) and single-audit interface, the predictive quotient is the unit sphere with its Borel structure, and the hypotheses of Theorem 3.1 hold with $A(R) = R$, $\kappa = 1$ and $B = k^2/\eta^2 - 1$. Theorems 4.3 and 5.4 therefore give the sequential and autonomous resource curves; on the circle Theorem 5.2 is exact. Suppressing reports gives exactly the spherical-atomic law and estimates of Theorem 6.1.*

Proof. The word kernel is $w \mapsto w \parallel R$ with law $h(dR)$. Thus no finite-dimensional manifold was assumed for the raw state. Nonnegativity and normalization of its induced coordinate kernel follow from the two factors and $|\eta x_I| \leq 1$. Conditional on a reported R , the coordinate functions pull back to Rx . Their squared integral matrix is I . The report law is independent of every entering point and all earlier histories, so the required conditional independence is an identity of the raw kernel, not a filtering approximation. Coordinate audits separate all points. Any finite continuation of reported rotations followed by the one audit remains affine in the entering coordinates, including report-dependent choices of the final audit, because the report law is state independent. Thus the finite test space closes. The Borel section described in Section 2 proves realized minimality.

A performed first Haar acquisition sends x_* to σ_d , and each subsequent acquisition preserves that law; this proves the actual acquired measure. The update is an isometry on the entire sphere and unit ball, and the report law has total-variation distance zero between different predictive points. The audit probabilities are Lipschitz in the coordinates. Thus global covering, small-ball mass and update stability are precisely the conditions proved in Lemma 4.2, not supplied model-specific exponents. Conditional on a hidden Haar transform the coordinate mean is zero, and subsequent visible rotations preserve it, proving the singular claim. All response tables and rotations used by the observer are independent of the actual hidden point. There are exactly $N = n + 2$ physical calls. $\square$

The full predictive quotient here refers to the declared future experiment, with one terminal linear audit. Adding repeated nonlinear observations changes the test space; it is not covered merely by retaining the same coordinates. Known orthogonal actions selected from the counted label do not improve the lower: conditioning on their fresh randomness leaves the reported Haar product isotropic. Actions that reveal the absolute point, undo an unknown transform from a free history tape, or reprepare the system are not in this interface.

7.2. A noncommuting quantum instrument. Let $\sigma_1, \sigma_2, \sigma_3$ be the Pauli matrices. Prepare

$$\rho_* = (I + rx_* \cdot \sigma)/2, \quad x_* \in \mathbb{S}^2, \quad 0 < r \leq 1.$$

The contrast r is a specified preparation parameter, not a value learned for free. For Haar probability h on $SU(2)$, the reported positive instrument is

$$(27) \quad \mathcal{I}(dU)(\rho) = U\rho U^* h(dU).$$

Each density map is completely positive, and the integrated instrument is trace preserving. Its report probability is $h(dU) \text{tr} \rho$. The audit chooses $I \in \{1, 2, 3\}$ uniformly and measures the two-outcome effect $(I + S\sigma_I)/2$. The scored vector is $Z = (3/r)Se_I$. No joint measurement of the three Pauli observables is asserted.

Theorem 7.2 (Noncommuting realization and rank change). *For the instrument (27), the test coordinate x in $\rho = (I + rx \cdot \sigma)/2$ is a realized predictive quotient for the declared instrument–audit interface. The same general energy and sphere theorems apply with $d = 2$, $\kappa = 1$, $V = 9/r^2$. The state family and the audit effects do not share an eigenbasis. Replacing the instrument by*

$$(28) \quad \mathcal{I}_t(dU)(\rho) = \alpha_t U \rho U^* h(dU), \quad \mathcal{I}_t(\{\perp\})(\rho) = (1 - \alpha_t) \text{tr}(\rho) I/2$$

verifies Theorem 6.1. For $r = 1$, the actual conditional states change from rank one to rank two on the first hidden report. The true rank-one mass is w_t , not one.

Proof. The identities $U(x \cdot \sigma)U^* = (R_U x) \cdot \sigma$ define the usual surjection from $SU(2)$ to $SO(3)$. The pushforward of Haar probability is Haar probability because it is normalized and invariant. The instrument therefore transports the centered test coordinates by $A(U) = R_U$ with state-independent reports and $\int A^T A dh = I$. The three single-query effects separate all Bloch coordinates; every allowed future continuation remains affine in ρ . A Borel choice of a unitary taking x_* to x on two coordinate charts gives a section. The audit calculation yields $\mathbb{E}[Z \mid \rho] = x$ and $\mathbb{E}\|Z\|^2 = 9/r^2$.

For two nonparallel vectors x, y , the commutator is proportional to $ir^2(x \times y) \cdot \sigma$, and is nonzero; thus a shared diagonal representation would be false. All continuity, mass and stability requirements nevertheless follow from the same spherical calculations, since the report probability does not depend on x . Haar twirling has $\int U \rho U^* dh = \text{tr}(\rho) I/2$, obtained by invariance and trace. The hidden branch of (28) is therefore the actual conditional state after an unreported unitary, not an artificial reset inserted by the observer. Once it occurs, later unitaries preserve $I/2$. This proves the mixed law and the rank statement, and hence all the singular inequalities. $\square$

The physical qubit is the experiment being predicted, not uncharged observer memory: the observer may neither store an ancillary quantum state nor perform extra measurements. Only its classical labels survive between reports. Completely positive finite-dimensional matrix identities are all that is used; no conclusion about unbounded-operator closures follows. Both realizations verify the hypotheses of the general theorem after that theorem has been proved, not conversely.

7.3. Higher-dimensional pure-state orbits: intrinsic rather than ambient dimension.

Let $D \geq 2$, V_D be the real Hilbert space of traceless Hermitian $D \times D$ matrices with the Hilbert–Schmidt norm, and $c_D = \sqrt{1 - 1/D}$. Prepare a known rank-one projector Π_* and use Haar conjugation on $SU(D)$. The centered coordinate is

$$X = (\Pi - I/D)/c_D, \quad \mathcal{O}_D = \{(\Pi - I/D)/c_D : \Pi \text{ rank one}\}.$$

Its norm is one, but its acquired dimension will be $d_D = 2(D - 1)$, not $\dim V_D = D^2 - 1$. These facts will follow from the raw unitary law.

Lemma 7.3 (Uniform mass across all conjugacy orbits). *For every $D \geq 2$ there are constants $C_D, r_D > 0$ such that for every traceless Hermitian V with $\|V\|_{\text{HS}} = 1$, every Hermitian center Z and $0 < r \leq r_D$,*

$$(29) \quad h\{U : \|UVU^* - Z\|_{\text{HS}} \leq r\} \leq C_D r^{2(D-1)}.$$

The pure-state orbit $\mathcal{O}_D$ has global covers of radius $C_D m^{-1/(2(D-1))}$. The stabilizer of any one of its points has fixed subspace exactly its line in V_D .

Proof. Write the eigenvalues of V in decreasing order. Their sum is zero and sum of squares is one, so their range is at least $1/\sqrt{D}$. Some consecutive gap is at least $g_D = 1/(\sqrt{D}(D - 1))$, say between ranks j and $j + 1$. For two matrices W, W' in this orbit, let P, P' be their top- j spectral projections. The variational characterization of the top eigenspace gives

$$\text{tr}(W(P - P')) \geq g_D(j - \text{tr}(PP')), \quad \text{tr}(W'(P' - P)) \geq g_D(j - \text{tr}(PP')).$$

Adding and using Cauchy–Schwarz yields $\|P - P'\|_{\text{HS}} \leq g_D^{-1} \|W - W'\|_{\text{HS}}$. If the ball in (29) meets the orbit, choose one W' in it; all other points in the ball have their top projections within $2r/g_D$ of P' . The random top projection is Haar on the complex Grassmannian of rank j .

For clarity its local mass estimate follows directly from Haar frames. Near a fixed rank- j plane use the graph chart $A \in \mathbb{C}^{(D-j) \times j}$. Normalizing a full-rank complex Gaussian frame gives Haar measure. Write that frame as $(B^T, C^T)^T$ with square invertible upper block B , and substitute $C = AB$. The real Jacobian is $|\det B|^{2(D-j)}$. Integrating $\exp[-\text{tr } B^*(I + A^*A)B] |\det B|^{2(D-j)}$ over B , then substituting $(I + A^*A)^{1/2}B$, gives density proportional to $\det(I + A^*A)^{-D}$. On a fixed neighborhood this density is bounded and projector distance is comparable to $\|A\|_{\text{HS}}$. Thus a radius- s ball has mass at most $C_{D,j} s^{2j(D-j)}$. Unitary invariance makes the constant independent of the center. Since $2j(D-j) \geq 2(D-1)$ and there are only $D-1$ possible ranks, the stated uniform bound follows. This proof does not assume a uniform simple spectrum.

For rank one, the same chart density is also bounded below locally and the chart has $2(D-1)$ real coordinates. Compactness, finitely many charts and a maximal separated set give the asserted global covers. Finally the stabilizer of a rank-one projector acts arbitrarily unitarily on its orthogonal complement (with the determinant corrected on the first line). Its fixed Hermitian matrices are scalar on each of the two blocks; the traceless condition leaves exactly the line through $\Pi - I/D$. $\square$

Theorem 7.4 (Projective quantum acquired-geometry law). *Prepare $(1-r)I/D + r\Pi_*$ with specified $0 < r \leq 1$ and use the raw instrument $\mathcal{I}(dU)(\rho) = U\rho U^*h(dU)$. Let $F_1, \dots, F_k$ be any Hilbert–Schmidt orthonormal basis of V_D , $k = D^2 - 1$. At audit draw I uniformly and measure the effects $(I + SF_I)/2$, which are positive since $\|F_I\|_{\text{op}} \leq 1$. Score $Z = (k/(rc_D))Se_I$. Then the hypotheses of Theorem 4.1 hold with*

$$d = 2(D-1), \quad V = k^2/(r^2 c_D^2), \quad B = V - 1.$$

In particular, for $M \geq n+1$,

$$(30) \quad \mathcal{R}_{\text{aut}}(n, M) - B \asymp_D \min\{1, n^{D/(D-1)}(M-1)^{-1/(D-1)}\},$$

whereas the checkpoint excess is of order $m^{-1/(D-1)}$. Independent suppressed reports verify Corollary 6.2; for $r = 1$ the conditional rank changes from one to D with actual masses $w_n, 1 - w_n$.

Proof. The centered raw audit mean is exactly the coordinate vector of $X = (\rho - I/D)/(rc_D)$, and its second moment is the displayed V . Conjugation acts orthogonally on V_D ; its only invariant traceless matrix is zero. The word of reported unitary instruments pulls back the finite coordinate tests linearly. As in the qubit proof, these tests separate the declared conditional states and have a Borel section, with pure-state projectors constructed from local normalized vectors. Haar conjugation derives the actual orbit measure. Lemma 7.3 verifies both the mass certificate for every possible incoming mean direction and the global cover and stabilizer certificates. This uniformity is crucial: incoming means need not themselves lie on the pure-state orbit. Apply the general theorem and substitute $d = 2(D-1)$ to obtain (30). Haar twirling sends every state to I/D , proving the hidden-report and rank assertions. The instrument family is noncommuting for nonorthogonal pure-state preparations, and no shared-basis reduction was used. $\square$

8. FIXED-START ACCESSIBILITY AND MARKED OCCUPATION

The nonreset theorem and the marked physical-time theorem describe different interfaces. We record their point of contact without replacing either. For a fixed finite retained boundary, let an excursion from row i have duration $\tau \geq 1$, exit label I_1 , and bounded per-call scores with complete reward R . Define

$$P_{ij} = \mathbb{P}_i(I_1 = j), \quad T_{ij} = \mathbb{E}_i[\tau \mathbf{1}_{I_1=j}], \quad r_i = \mathbb{E}_i R.$$

Let g be the classwise physical-time gain, with absorption weights from the actual entering row, and put $b = r - Tg$. Duration and exit are not replaced by independent marks. For a family X with uniformly bounded all-row mean return $\bar{\mu}$ define

$$k_N(x) = \inf_u \{ \|b_x - (I - P_x)u\|_\infty + \text{osc}(u)/N \},$$

$$\Delta_N(X) = \sup_{x \in X} \max_{1 \leq t \leq N} \|H_t(x)\|_\infty / N, \quad H_t = t(J_t - g).$$

The marked renewal inequality, with its physical-return remainder $B_N(X)$, is

$$(31) \quad \sup_X k_N \leq (\bar{\mu} + 2)\Delta_N(X) + B_N(X).$$

The reverse bound $\Delta_N(X) \leq \sup_X k_N + B_N(X)$ is also valid. Both proofs, including the actual first-excursion renewal converse, are retained in the complete marked-occupation companion. The remainder always belongs to the family over which the supremum is taken, including a product family of occupation tests. The exact finite dual of k_N is

$$(32) \quad k_N(x) = \max \left\{ z^T b_x : \|z\|_1 \leq 1, \quad \frac{1}{2} \|(I - P_x)^T z\|_1 \leq N^{-1} \right\}.$$

The vector z is signed, may depend on x , and is an analytical certificate rather than a runtime input or an occupation probability.

Proposition 8.1 (Observable access from one prescribed start). *Suppose a parameter-independent paid initialization of at most L calls, padded to L if necessary, produces a boundary row I and reports its tag, with probabilities $\rho_i(x) \geq a > 0$ for all rows, all $x \in X$. Conditional on that tag the subsequent excursion law is exactly the declared law from that row. Assume the test family includes the gated bounded-score tests $\mathbf{1}_{I=i}$ times the row- i score; initialization calls are unscored. Define the discrepancy of this accessible test family by*

$$\Delta_N^{\text{vis}}(X) = \sup_{x \in X} \max_{i, 1 \leq t \leq N} \frac{\rho_i(x) |H_t(x)_i|}{N + L}.$$

For every fixed x and every feasible z in (32),

$$(33) \quad \Delta_N^{\text{vis}}(X) \geq \frac{aN}{(N + L)(\bar{\mu} + 2)} (z^T b_x - B_N(X))_+.$$

Centering by the limiting gain is an analytical comparison; runtime gates only the raw score and never reads g , x or z . This is a fixed-initial-law, observable-test lower, not a lower for an ungated scalar average or for an arbitrary decision loss.

Proof. Disintegration at the initialization tag gives exactly the contribution $\rho_i(x)H_t(x)_i$ for a gated test. Hence $\Delta_N^{\text{vis}}(X) \geq aN\Delta_N(X)/(N + L)$. For the fixed x, z , dual feasibility and (31) give $z^T b_x \leq (\bar{\mu} + 2)\Delta_N(X) + B_N(X)$. Combine the inequalities and nonnegativity. The tag gate requires a stored Boolean value once a particular test is chosen. An autonomous tester additionally pays for its prefix clock, for example by a product with $2(N + L + 1)$ labels. The physical initialization costs L calls. These are extra resources of the accessibility interface, not free properties of the original observer. $\square$

Accessibility alone does not prevent scalar cancellation. Let P exchange two rows, $\tau = 1$, and row rewards be $0, 1$. Then $g = (1/2, 1/2)$, $b = (-1/2, 1/2)$, and $H_t = b$ for odd t , $H_t = 0$ for even t . Thus for every $N \geq 1$, $\Delta_N = 1/(2N)$ and the dual (or the one-variable primal) gives $k_N = 1/(2N)$. From the fixed uniform initial law the ungated scalar discrepancy is zero at every prefix, although both rows are accessible. The reported tag restores the gated test. If a row has zero initialization probability, an obstruction confined to its inaccessible closed class need not be visible at all. Neither failure is remedied by calling a signed dual vector an acquired measure.

Lemma 8.2 (Physical crossing convention). *For positive integer durations set $T_0 = 0$, $T_j = \sum_{\ell=1}^j \tau_\ell$ and $C_N = \min\{j \geq 1 : T_j \geq N\}$. Then $C_N \leq N$; excursion j starts before call N exactly when $T_{j-1} < N$, and there are C_N such starts. In the older shifted convention $S_k = T_{k-1}$ and $K_N = \min\{k \geq 1 : S_k \geq N\}$, one has $K_N = C_N + 1 \leq N + 1$, while the number of summands $k < K_N$ is at most N .*

Proof. Since $T_j \geq j$, the crossing occurs by $j = N$. Strict increase gives the assertion about starts, including when $T_{C_N} = N$: the next excursion starts at N , not before it. Substituting $S_k = T_{k-1}$ proves the shifted identities. The overshoot is $T_{C_N} - N$. Stopped sums use the predictable indicator $\mathbf{1}_{T_{j-1} < N}$, not an independence claim about the final excursion mark. $\square$

The complete retained physical-time proof uses this corrected convention; its estimate is unchanged. The bounded-Lipschitz test class for subprobability occupations is $\{f : \|f\|_\infty \leq 1, \text{Lip}(f) \leq 1\}$, so it includes the constant one and controls scored mass as well as location. For the resolution test $\phi = \min(\text{dist}(\cdot, C)^2, r^2)$ with $0 < r \leq 1$, $\phi/(2r)$ belongs to this class. Projection of centers into $\text{conv } K$ cannot increase distance to points of K . Repeated readouts at different control labels yield at most, not necessarily exactly, M distinct codepoints. These conventions apply to the retained geometric lower without promoting a fixed-exploration witness to an all-controller witness.

9. RESOURCE-CERTIFIED COMPOSITION AND IMPLEMENTATION

A task-compatible causal morphism has a parameter-independent report map, a strategy lift, a counted internal state, a physical clock map and a readout map. The action ports and the report deadline must agree. The environment-owned final audit remains the same physical audit: it cannot be regenerated by the simulator. Total variation below uses $\text{TV}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$. For bounded loss in $[0, L]$, an error ϵ on the complete report–response–audit law changes its expectation by at most $L\epsilon$.

Proposition 9.1 (Finite-horizon composition). *Suppose a simulator with S internal labels transforms a source experiment into a target at complete task-law discrepancy at most ϵ , uniformly over the declared target strategies and fixed unknown parameters. A target observer with M labels lifts to a source observer with at most MS labels. If it uses at most n virtual acquisitions, each using at most c physical calls, the lifted count is at most cn plus the explicitly charged setup and audit. Suppose an implemented report/readout mechanism is further coupled to the ideal one with probability of any decision mismatch at most ζ , and its terminal readout error on the agreement event is at most γ . If $\|Z\| \leq K$ and ideal and implemented readouts lie in the radius- A ball, then*

$$(34) \quad R_{\text{source}} \leq R_{\text{target}} + (K + A)^2(\epsilon + \zeta) + 2(K + A)\gamma + \gamma^2.$$

The same comparison reverses only when a reverse task-compatible simulator is supplied. Iterated simulator errors add by the triangle inequality, while their retained states multiply. Program description, temporary workspace and execution time add with the stated call schedule rather than being identified with $\log_2(MS)$.

Proof. Retain the pair of observer and simulator states. The causal strategy lift uses only this pair and the current source report; its joint transitions and readout give the target law up to ϵ . Couple the further implementation with its ideal run until the first mismatch. On a mismatch the squared loss differs by at most $(K + A)^2$; on agreement use $|\|Z - a - e\|^2 - \|Z - a\|^2| \leq 2(K + A)\|e\| + \|e\|^2$. Integration proves (34). No infinite-horizon or unbounded-stopping limit is taken. Products of state sets and concatenation of programs prove the resource assertions. $\square$

Proposition 9.2 (A charged finite-bit circle implementation). *For the circle program of Theorem 5.4, let $K = M - 1 = \sum_t m_t$, and suppose the acquisition device returns an angle approximation*

with circular error at most $h = 2^{-p}$ at cost $T_{\text{rep}}(p)$ and temporary space $W_{\text{rep}}(p)$. All approximation routines and their worst-case error bounds are part of the program specification. There is a finite-bit implementation with the same M persistent labels, program description $O(M(p + \log M) + n \log M)$ bits, temporary workspace $O(p + \log M) + W_{\text{rep}}(p)$ bits, and online time $O(n(p + \log M)^2) + nT_{\text{rep}}(p)$, whose risk exceeds the ideal value by at most

$$(35) \quad C(K_Z + A)^2 \min(1, (M - 1)h) + C(K_Z + A)nh + Cn^2h^2.$$

Here $\|Z\| \leq K_Z$ and A bounds the readouts. The construction is a finite-precision approximation, not an exact finite-bit evaluation of arbitrary real report boundaries.

Proof. Store rational approximations to each phase's regular-arc endpoints, output directions and coefficients a_m to absolute accuracy at most a fixed multiple of h ; fixed-precision arithmetic evaluates the phase's rotated old direction with angle error at most Ch . Constants can be compiled by convergent sine/cosine series with certified remainders. Reading the table and multiplying/dividing integers of $O(p + \log M)$ bits has the displayed schoolbook bounds; the compilation cost is separate from online time and is finite under these explicit routines.

Couple exact and rounded executions by the same real reports. Until their first state mismatch the incoming label is identical. The new exact angle is uniform conditional on that label, so it falls within Ch of one of the m_t boundaries with probability at most Cm_th . A union bound over the calls gives $\zeta \leq \min(1, CKh)$. On the agreement event, the product readout coefficient and direction have error at most Cnh , since all exact factors lie in $[0, 1]$ and approximations can be clipped there. Apply Proposition 9.1. The phase index and the old direction are encoded by the retained label; workspace is erased between calls. Report precision is a paid input capability, and the audit is still unavailable until after commitment. $\square$

We next relate actual disclosure loss to a causal deficiency; the audit condition is essential. Let $\epsilon_*(\alpha)$ be the infimum, over all causal simulators with arbitrary internal memory but no extra physical acquisition, of the total variation between (i) the ideal disclosed rotation reports with the original terminal audit and (ii) simulated reports from the erased experiment with that same physical audit. Reports must be emitted at their original deadlines. The simulator may not change the prepared system, the physical rotations or the audit; it only transforms the visible reports using fresh randomness. This is a specified task-compatible deficiency, not an unrestricted statistical comparison permitting a new independent audit.

Proposition 9.3 (Attained order of disclosure deficiency). *In the classical realization with contrast η and $k = d + 1$, in the qubit realization with $k = 3$ and contrast r , and in the projective realization with $k = D^2 - 1$, write $\chi = \eta$, $\chi = r$, or $\chi = rc_D$, respectively. For all $\alpha_t \in [0, 1]$,*

$$(36) \quad \frac{\chi}{2k}(1 - w_n) \leq \epsilon_*(\alpha) \leq 1 - w_n.$$

The upper uses a stateless report transformation which replaces each hidden report by a fresh Haar draw. Consequently, when phase report laws are mutually equivalent and $M \geq 2n + 1$, the internally clocked same-task minimax risk with the invisible calibration coordinate satisfies

$$(37) \quad \mathcal{R}_{\text{aut}}^{\text{er}}(N, M, \delta) - B \asymp_{d, \chi} \min \left(1, \frac{(N - 2)^{1+2/d}}{(M - 1)^{2/d}} \right) + \delta^2 + \epsilon_*(\alpha), \quad N = n + 2.$$

The exact surviving mass remains in Theorem 6.1; the comparable sum does not replace it in that theorem.

Proof. For the upper couple the physical system, all rotations, masks and audit. On the event of no erasure the report transformation equals the ideal transcript, an event of probability w_n . On its complement any two complete laws differ in total variation by at most its probability. Fresh Haar replacement respects deadlines and needs no persistent simulator label.

For the lower, let $\tilde{x}$ be the unit point obtained by multiplying the simulated reported rotations into the known prepared point. Consider the bounded statistic $S\tilde{x}_I \in [-1, 1]$ of the reported transcript and the subsequently revealed audit. Under the ideal experiment its expectation is χ/k . In the erased experiment, conditional on the entire allowed visible history before the audit, the predictive coordinate is zero on any hidden-transform history and has norm one on the surviving history. Simulator randomness is independent of the physical state conditional on that information. Since the query I is drawn only at audit, its expectation under any causal simulator is at most $w_n\chi/k$. The expectation gap is at least $(1 - w_n)\chi/k$, and the total-variation bound for a statistic in $[-1, 1]$ is twice its total variation. This proves the lower even for unbounded simulator memory. In the quantum case the same argument uses the conditional density I/D on the erased event, not an assumed hidden joint value of incompatible measurements.

For (37) combine the homogeneous (or circle) clock bound, the singular upper using a counted inactive label in every phase, and the calibration corollary. In the range $M - 1 \geq 2n$, balancing the widths leaves at least one active label per phase and changes the geometric rate by only dimension-dependent constants. The elementary inequality $\frac{1}{2}(a + 1 - w) \leq 1 - w + wa \leq a + 1 - w$ for $0 \leq a, w \leq 1$ then replaces the surviving-mass expression by the comparable sum. Finally use (36). No allowed error tolerance is automatically treated as a necessary risk floor: the lower uses this actual erased interface and its protected audit. $\square$

10. POSITION OF THE RESULTS AND MATHEMATICAL SCOPE

The conditional-square decomposition and the static quantization identity are standard, and the circle quantizer is classical [1, 2]. The new assertion is not that conditional expectation loses variance. It is the identification of the additional pooled-row obstruction, the exact condition under which every rowwise loss can be attained by one recursively reused labeling, and its consequence for a nonreset relative-report experiment with fully counted deadline memory. The spherical law and the exact balanced circle formula quantify a distinction invisible to the one-time acquired distribution: identical spherical marginals coexist with a horizon-dependent causal cost. The two positive realizations identify the same mechanism without a shared classical eigenbasis or an assumed contracting filter.

The nearest comparison in causal coding is the zero-delay theory for Markov sources. Linder and Yüksel [3] establish structural and existence results for optimal quantizers, including finite-horizon deterministic convex-cell quantizers under their hypotheses. Ghomi, Linder and Yüksel [4] obtain stationary optimal coding and near-optimal finite-memory codes for linear vector sources. In those models the encoder observes the current source. Here the device sees a relative instrument report and a finite retained label, not the current source point. Observing that point at every call would allow a fresh static quantizer and remove the product obstruction. Our internally charged deadline also differs from a publicly scheduled coding architecture. The conditional expectation tools are shared; neither observation interface is a special notation for the other.

Strong uniform values for partially observed control [5] concern one strategy that is near optimal over sufficiently long horizons and a long-run criterion. Finite-memory near-optimality in long-run POMDPs [6] permits the memory needed for a prescribed accuracy to be chosen. Yu and Bertsekas [7] establish near optimality of finite-state controllers under the stated average-cost hypotheses. Theorems 3.1 and 5.4 instead fix the complete finite-state resource and the terminal test, quantify all finite horizons, and include the internal deadline. They neither imply a general strong uniform value nor contradict existence of increasingly large nearly optimal controllers. The tracking horizon here is not a claim of decreasing static estimation risk with increasing sample count.

History-space existence and finite-window approximation [8], filter-contraction average-cost approximation [9], and strategic-measure optimality [10] address substantially broader control or

measurability questions. Our raw transport condition is narrower and explicitly verifiable; within it the full-history update is an isometry rather than a strict contraction. General state-dependent report likelihoods are not covered by writing a nonlinear filter as a coordinate map. Multichain policy existence and computation [11] likewise concern a distinct optimization object from the exact integer allocation solved here. The allocation corollary supplies a genuine arithmetic complexity bound for this class, not a polynomial-time global synthesis theorem for POMDPs.

For the retained physical-time theory, ergodic approximation functionals [12] and Poisson representations [13] are classical ingredients. Its contribution remains the marked duration–exit converse with a maximal-prefix normalization, the indispensable physical-return remainder, the finite signed dual, and the resolution-localized transfer of actual scored occupation. These are three separate layers: a quantitative inequality for a fixed finite marked boundary, a compact-family qualitative/robust theorem under its return hypotheses, and a spatial matching theorem requiring actual-mass witnesses plus implementable covers. Proposition 8.1 adds a prescribed-start test transfer; it does not make those hypotheses disappear. The nonreset energy theorem is independent of that renewal proof.

The present compatibility condition is necessary and sufficient for saturating the local energy coefficient in the declared finite test interface. We have not proved that it characterizes every finite-memory causal experiment, that the spherical lower product is attained on every higher-dimensional orbit, or that arbitrary acquired measures admit compatible recursive quantizers. The matching spherical rates do not require that stronger claim. Arbitrary unknown kernels, nondominated common-program compactness, unrestricted sensing controls, unbounded stopping, infinite-horizon optimality, and a fully matched program/workspace/precision/time/minimal-simulator region require additional mathematics. These are specific scope boundaries, not replacements of the general acquired-geometry problem. The results proved above retain that problem’s causal and resource content while supplying an exact nonreset criterion and a sharp resource law on a nontrivial raw experiment class.

REFERENCES

- [1] S. Graf and H. Luschgy, *Foundations of Quantization for Probability Distributions*, Lecture Notes in Mathematics 1730, Springer, 2000.
- [2] J. Rosenblatt and M. K. Roychowdhury, *Uniform distributions on curves and quantization*, arXiv:1809.08364.
- [3] T. Linder and S. Yüksel, *On optimal zero-delay coding of vector Markov sources*, IEEE Trans. Inform. Theory **60** (2014), 5975–5991, doi:10.1109/TIT.2014.2346780.
- [4] M. Ghomi, T. Linder and S. Yüksel, *Zero-delay lossy coding of linear vector Markov sources: optimality of stationary codes and near optimality of finite memory codes*, arXiv:2103.10810.
- [5] X. Venel and B. Ziliotto, *Strong uniform value in gambling houses and partially observable Markov decision processes*, SIAM J. Control Optim. **54** (2016), 1983–2008, doi:10.1137/15M1043340.
- [6] K. Chatterjee, R. Saona and B. Ziliotto, *Finite-memory strategies in POMDPs with long-run average objectives*, Math. Oper. Res. **47** (2022), 100–119, doi:10.1287/moor.2020.1116.
- [7] H. Yu and D. P. Bertsekas, *On near optimality of the set of finite-state controllers for average cost POMDP*, Math. Oper. Res. **33** (2008), 1–11, doi:10.1287/moor.1070.0279.
- [8] S. Yüksel, *Another look at partially observed optimal stochastic control: existence, ergodicity, and approximations without belief-reduction*, arXiv:2301.11244.
- [9] Y. E. Demirci, A. D. Kara and S. Yüksel, *Average cost optimality of partially observed MDPs: contraction of nonlinear filters and existence of optimal solutions and approximations*, SIAM J. Control Optim. **62** (2024), 2859–2883, doi:10.1137/24M1643736.
- [10] H. Yu, *On strategic measures and optimality properties in discrete-time stochastic control with universally measurable policies*, Math. Oper. Res. **49** (2024), 1734–1760, doi:10.1287/moor.2022.0188.
- [11] A. Leizarowitz, *An algorithm to identify and compute average optimal policies in multichain Markov decision processes*, Math. Oper. Res. **28** (2003), 553–586, doi:10.1287/moor.28.3.553.16388.
- [12] S.-Y. Shaw, *Non-optimal rates of ergodic limits and approximate solutions*, J. Approx. Theory **94** (1998), 285–299, doi:10.1006/jath.1998.3186.
- [13] P. W. Glynn and A. Infanger, *Solution representations for Poisson’s equation, martingale structure, and the Markov chain central limit theorem*, Stochastic Systems **14** (2024), 47–68, doi:10.1287/stsy.2022.0001.